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title: "Week 8 Problem Set"
author: "Ava https://github.com/avaelisa/soc10problemsets-ava/blob/main/Week8PSetAva.Rmd"
date: "5/26/26"
output: pdf_document
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```

[Note: the scoring guidelines in brackets can be deleted before submitting. Overall presentation/cleanliness of document (e.g. writing and code presented clearly, no unnecessary messages/warnings) 1 point.]

1. Start a new document, `problemset8.Rmd`, in your `soc10problemsets` repository. (You can download a template [here](https://github.com/Soc10Spring2026/soc10problemsets_templates), but be sure to save it in your own `soc10problemsets` repo). Use the Guardian API to download news stories related to the the World Cup. Put the data in `'tidytext'` format (each row should have a single word). __Commit your changes__.

```
```{r libraries, echo = T, message=F, warning=F}
library(guardianapi)
library(dplyr)
library(tidytext)
```

```
#my_key <- "524effbe-627c-4e19-91f5-f02d72877c45"
gu_api_key()
```

```
...
```

[Successfully download stories using the Guardian API (1 point). Gather relevant data (0.5 points) and put it in `tidytext` format (0.5 points).]

```
```{r q1, echo = T}
```

```
guardian <- gu_content("World Cup",
                      from_date = "2025-01-1",
                      to_date = "2026-05-11")
```

```
guardian %>%
  select(headline, byline, web_publication_date, body_text)
```

```
#View(guardian)
```

```
tidy_guardian <- guardian %>%
  unnest_tokens(word, body_text) %>%
  anti_join(stop_words)
```

```
head(tidy_guardian)
```

```
#look at words
tidy_guardian %>%
  select(word)
```

```
...
```

2. Use `'RedditExtractoR'` to download Reddit comments (or threads) for a topic of interest. Explain how you picked the topic and what we might learn from it. Be sure that you download data from multiple subreddits, and describe each briefly. __Commit your changes__.

I selected topics that relate to law enforcement on reddit. I selected these as I knew they existed and related to my topic of interest. The subreddits that I selected were "AskLE" or ask law enforcement and "ProtectandServe". Ask Law Enforcement can provide

insight into law enforcement related inquiries. We can learn what questions civilians have for law enforcement officers as well as the answers to some of these questions; however, one should proceed with some caution in regard to the accuracy of the answers. Protect and serve can provide insight into conversations surrounding law enforcement and the law enforcement community. We can learn about any questions or concerns or general thoughts that law enforcement officers have. In addition, we can also learn how they talk about their work and view their work. I have downloaded comments from these subreddit pages as well as related information to the comments i.e. url, author name, etc.

```
```{r q2, echo = T, message = FALSE, warning= FALSE}

library(RedditExtractoR)

#??find_subreddits

we can get urls of the law enforcement subreddits (protect and serve, ask LE)
askLE_urls <- find_thread_urls(subreddit = "AskLE",
 period = "day")

PAS_urls <- find_thread_urls(subreddit = "ProtectandServe",
 period = "day")

askLE_urls <- as.data.frame(askLE_urls)
PAS_urls <- as.data.frame(PAS_urls)

extract comments from these pages
askLE_comments <- get_thread_content(askLE_urls$url)
PAS_comments <- get_thread_content(PAS_urls$url)

take a look
askLE_comments$threads
askLE_comments$comments

PAS_comments$threads
PAS_comments$comments

#code for alternatives, not using these
#extract law enforcement subreddits
#law_enforcement_subreddits <- RedditExtractoR::find_subreddits("police")
#alternatively, we can find urls of all pages related to police
#police_urls <- RedditExtractoR::find_thread_urls(keywords = "police")

```
```

[Successfully download Reddit comments (1 point) from multiple subreddits (0.5 points). Describe your data accurately (0.5 points).]

3. Choose two subreddits, and compare the frequencies of words between them. (Similar to what is done in Figure 8.3). __Commit your changes__.

```
```{r q3, echo = T, message = FALSE, warning = FALSE}

#put into tidy text and filter for word

askLE_comments2 <- askLE_comments$comments %>%
 mutate(type = "askLE")

PAS_comments2 <- PAS_comments$comments %>%
 mutate(type = "ProtectandServe")

askLE_comments3 <- askLE_comments2 %>%
 unnest_tokens(word, comment) %>%
```

```

anti_join(stop_words) %>%
 select(word, type)

head(askLE_comments3)

PAS_comments3 <- PAS_comments2 %>%
 unnest_tokens(word, comment) %>%
 anti_join(stop_words) %>%
 select(word, type)

head(PAS_comments3)

#frequency
askLE_comments3 %>%
 count(word, sort = TRUE)

PAS_comments3 %>%
 count(word, sort = TRUE)

library(tidyr)
frequency <- bind_rows(askLE_comments3,
 PAS_comments3) %>%
 count(word, type) %>%
 group_by(type) %>%
 mutate(proportion = n / sum(n)) %>%
 select(-n) %>%
 pivot_wider(names_from = type, values_from = proportion)

frequency %>%
 head()

library(scales)
library(ggplot2)

expect a warning about rows with missing values being removed
ggplot(frequency, aes(x = askLE, y = ProtectandServe,
 color = abs(askLE - ProtectandServe))) +
 geom_abline(color = "gray40", lty = 2) +
 geom_jitter(alpha = 0.1, size = 2.5, width = 0.3, height = 0.3) +
 geom_text(aes(label = word), check_overlap = TRUE, vjust = 1.5) +
 scale_x_log10(labels = percent_format()) +
 scale_y_log10(labels = percent_format()) +
 scale_color_gradient(limits = c(0, 0.001),
 low = "darkslategray4", high = "gray75") +
 theme(legend.position="none") +
 labs(x = "AskLawEnforcement", y = "ProtectandServe")

```

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[Create a plot to show term frequencies across two subreddits (1 point). The plot should be accurately labeled and presented (0.5 points).]

4. Create a plot comparing sentiment across two or more subreddits. (Similar to what is done in Figure 8.5). Commit your changes.

The sentiment that I selected is "NRC" from the tidytext package. NRC categorizes thousands of words based on the sentiments associated with them. The categories include positive, negative, anger, anticipation, disgust, fear, joy, sadness, surprise, and trust. I selected NRC because I appreciate that it is not a numeric scale (such as AFINN) and that it provides more detail into the sentiments beyond positive and negative (such as Bing). Further, NRC is frequently used in sentiment analysis. The graph below reveals the frequency of each of the words per each category identified by NRC.

```

````{r q4, echo = T, message = FALSE, warning = FALSE}

library(tidytext)
#install.packages("textdata")
library(textdata)
library(magrittr)
library(ggplot2)

look at nrc lexicon
nrc1 <- get_sentiments("nrc")

#ProtectandServe Graph
#join nrc with PAS comments
PAS_comments4 <- PAS_comments3 %>%
 inner_join(nrc1)

#take a look
PAS_comments4 %>%
 select(word, sentiment)

#in table form
table(PAS_comments4$sentiment)

ggplot(PAS_comments4, aes(y = sentiment))+
 geom_bar(aes(fill = sentiment))+
 theme_minimal()+
 labs(title = "Sentiments in Comments about ProtectandServe subreddit")

#AskLE Graph
#join nrc with askLE comments
askLE_comments4 <- askLE_comments3 %>%
 inner_join(nrc1)

#take a look
askLE_comments4 %>%
 select(word, sentiment)

#in table form
table(askLE_comments4$sentiment)

ggplot(askLE_comments4, aes(y = sentiment))+
 geom_bar(aes(fill = sentiment))+
 theme_minimal()+
 labs(title = "Sentiments in Comments about askLawEnforcement subreddit")

#comparison graph
law_enforcement <- bind_rows(askLE_comments3,
 PAS_comments3)

law_enforcement1 <- law_enforcement %>%
 inner_join(nrc1)

#take a look
law_enforcement1 %>%
 select(word, sentiment)

#in table form
table(law_enforcement1$sentiment)

ggplot(law_enforcement1, aes(y = sentiment))+
 geom_bar(aes(fill = sentiment))+

```

```
theme_minimal()+
labs(title = "Sentiments in Articles about Law Enforcement Subreddits")+
facet_wrap(~type, scales = "free_x")
```

```

[Use a sentiment dictionary to attach sentiments to your data (1 point). Comment on the sentiment chosen (0.5 points). Accurately present and label the plot (0.5 points).]

5. Which of the "families" of text analysis do your analyses from the prior questions work fall under? Describe how you might incorporate methods from the other families in your analysis. __Commit your changes__.

My data is part of the term frequency family and closed vocabulary family.

Term frequency shows "text as observations that vary in how often certain strings of characters (i.e. words) appear." This means that it shows how often a specific string of characters is used. In this graph, it shows how often a word is used in a subreddit comment section.

Closed vocabulary is based on pre-existing theories and a measure of text analysis based on a defined/established set of words. Since the NRC relies on an existing and pre-made word dictionary with preassigned sentiments, it falls under this category.

Open vocabulary is where a model is able to recognize and generate concepts using natural language/correlate each of K most common words with outcome variables. It generates new theories. This could be used in the policing subreddit to see if there are any correlations/other sentiments for words. For example, taser is one of the words used. Maybe for law enforcement officers, tasers or other weapons have a positive sentiment because they are used for protection. This is when having insight and flexibility is ideal. It can be used for future research to generate new theories/new information on sentiments based on conversations surrounding policing.

[Accurately describe the families of text analysis (0.5 points), which apply to PSet analyses (0.5 points), and how others might be useful for future research (0.5 points).]